

The accuracy is:

Scegli un'alternativa:

- ☒ a. The number of correct significant digits in approximating some quantity.
- ☐ b. The number of digits with which a number is expressed.
- ☐ c. None of the above.



La risposta corretta è: The number of correct significant digits in approximating some quantity.

Given two random variables X and Y , Bayes Theorem implies that $p(y|x) = \frac{p(x|y)p(y)}{p(x)}$ where:

Scegli un'alternativa:

- ☐ a. $p(x|y)$ is called prior distribution on x .
- ☐ b. $p(x|y)$ is called posterior distribution on y .
- ☒ c. $p(x|y)$ is called likelihood on y .



La risposta corretta è: $p(x|y)$ is called likelihood on y .

Given two random variables X and Y , Bayes Theorem implies that

$$p(y|x) = \frac{p(x|y)p(y)}{p(x)} \text{ where:}$$

Scegli un'alternativa:

- ☐ a. $p(y|x)$ is called prior distribution on x .
- ☐ b. $p(y|x)$ is called likelihood on y .
- ☒ c. $p(y|x)$ is called posterior distribution on y .



La risposta corretta è: $p(y|x)$ is called posterior distribution on y .

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1 e^{x_2}$, $g : \mathbb{R} \rightarrow \mathbb{R}^2$, $g(t) = (e^t, t)$, then, if $h(t) = f(g(t))$:

Scegli un'alternativa:

- ☐ a. $h'(t) = te^t$.
- ☐ b. $h'(t) = 2e^{2t}$.
- ☒ c. $h'(t) = e^{2t}(t + 1)$.



La risposta corretta è: $h'(t) = e^{2t}(t + 1)$.

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1^2 + x_1x_2$, $g : \mathbb{R} \rightarrow \mathbb{R}^2$, $g(t) = (\sin(t), \cos(t))$,
then, if $h(t) = f(g(t))$:

Scegli un'alternativa:

- ☐ a. $h'(t) = \sin(2t) - \sin^2(t)$.
- ☐ b. $h'(t) = \sin(t) - \sin^2(2t)$.
- ☒ c. $h'(t) = \sin(t) \cos(t) - \sin^2(t)$.



La risposta corretta è: $h'(t) = \sin(2t) - \sin^2(t)$.

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1^2 + x_2^2$, $g : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $g(x_1, x_2) = (x_2, x_1)$,
then, if $h(x_1, x_2) = f(g(x_1, x_2))$:

Scegli un'alternativa:

- ☒ a. $\nabla h(x_1, x_2) = (2x_1, 2x_2)$.
- ☐ b. $\nabla h(x_1, x_2) = (2x_2, 2x_1)$.
- ☐ c. $\nabla h(x_1, x_2) = (1, 1)$.



La risposta corretta è: $\nabla h(x_1, x_2) = (2x_1, 2x_2)$.

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1 e^{x_2}$, $g : \mathbb{R} \rightarrow \mathbb{R}^2$, $g(t) = (t, \log t)$, then, if $h(t) = f(g(t))$:

Scegli un'alternativa:

- ☐ a. $h'(t) = t + 1$.
- ☐ b. $h'(t) = t^2 + 1$.
- ☐ c. $h'(t) = 2t$.

La risposta corretta è: $h'(t) = 2t$.

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1^2 + x_1x_2$, $g : \mathbb{R} \rightarrow \mathbb{R}^2$, $g(t) = (t^2, t)$, then, if $h(t) = f(g(t))$:

Scegli un'alternativa:

- ☐ a. $h'(t) = t(2t - 1)^2 + t$.
- ☐ b. $h'(t) = 4t^2 + 2t + 1$.
- ☐ c. $h'(t) = t(2t + 1)^2 - 2t^2$.

La risposta corretta è: $h'(t) = t(2t + 1)^2 - 2t^2$.

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1^2 + x_2^2$, $g : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $g(x_1, x_2) = (x_1 e^{x_2}, x_2)$,
then, if $h(x_1, x_2) = f(g(x_1, x_2))$:

Scegli un'alternativa:

- ☐ a. $\nabla h(x_1, x_2) = (2x_1 e^{x_2} (e^{x_2} + x_1), 2e^{x_2})$.
- ☐ b. $\nabla h(x_1, x_2) = (2x_1 e^{2x_2} (e^{x_1} + x_1), 2x_2)$.
- ☒ c. $\nabla h(x_1, x_2) = (2x_1 e^{x_2} (e^{x_2} + x_1), 2x_2)$.



La risposta corretta è: $\nabla h(x_1, x_2) = (2x_1 e^{x_2} (e^{x_2} + x_1), 2x_2)$.

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1 x_2$, $g : \mathbb{R} \rightarrow \mathbb{R}^2$, $g(t) = (t, t^2)$, then, if $h(t) = f(g(t))$:

Scegli un'alternativa:

- ☒ a. $h'(t) = 3t^2$.
- ☐ b. $h'(t) = 3t^3$.
- ☐ c. $h'(t) = t^2$.



La risposta corretta è: $h'(t) = 3t^2$.

If

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

then:

Scegli un'alternativa:

- ☒ a. $K_2(A) = 1$.
- ☐ b. $K_2(A) = 4$.
- ☐ c. $K_2(A) = \frac{1}{2}$.



La risposta corretta è: $K_2(A) = 1$.

If

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

then:

Scegli un'alternativa:

- ☐ a. $K_2(A) = 4$.
- ☒ b. $K_2(A) = 2$.
- ☐ c. $K_2(A) = \frac{4}{3}$.



La risposta corretta è: $K_2(A) = 2$.

If vector $v = (10^6, 0)^T$ is approximated by vector $\tilde{v} = (999996, 1)^T$, then in $\|\cdot\|_2$ the relative error between v and $\tilde{v}$ is:

Scegli un'alternativa:

- ☐ a. $\sqrt{17} \cdot 10^{-6}$.
- ☐ b. None of the above.
- ☒ c. $4 \cdot 10^{-6}$.



La risposta corretta è: $\sqrt{17} \cdot 10^{-6}$.

If

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

then:

Scegli un'alternativa:

- ☐ a. $K_2(A) = \frac{1}{2}$.
- ☐ b. $K_2(A) = 4$.
- ☒ c. $K_2(A) = 2$.

✗

La risposta corretta è: $K_2(A) = 4$.

A random variable $X : \Omega \rightarrow \mathcal{T}$ is continuous when:

Scegli un'alternativa:

- ☐ a. $\mathcal{T}$ is countable.
- ☒ b. $\mathcal{T} = \mathbb{R}$.
- ☐ c. Ω is continuous.



La risposta corretta è: $\mathcal{T} = \mathbb{R}$.

A random variable $X : \Omega \rightarrow \mathcal{T}$ is discrete when:

Scegli un'alternativa:

- ☐ a. $\mathcal{T} = \mathbb{R}$.
- ☒ b. Ω is countable.
- ☐ c. $\mathcal{T}$ is countable.



La risposta corretta è: $\mathcal{T}$ is countable.

If

$$A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☐ a. $x = (1, 2)^T$ is an eigenvector of A .
- ☒ b. $x = (2, 1)^T$ is an eigenvector of A .
- ☐ c. $x = (0, 0)^T$ is an eigenvector of A .



La risposta corretta è: $x = (2, 1)^T$ is an eigenvector of A .

If

$$A = \begin{bmatrix} 4 & 2 \\ 1 & 3 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☐ a. $\lambda = 5$ is the eigenvalue associated with the eigenvector $x = (2, 1)^T$.
- ☒ b. $\lambda = 2$ is the eigenvalue associated with the eigenvector $x = (2, 1)^T$.
- ☐ c. $\lambda = 2$ is the eigenvalue associated with the eigenvector $x = (1, 2)^T$.



La risposta corretta è: $\lambda = 2$ is the eigenvalue associated with the eigenvector $x = (2, 1)^T$.

If

$$A = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☒ a. $x = (1, 1, 0)^T$ is an eigenvector of A .
- ☐ b. $x = (0, 1, 0)^T$ is an eigenvector of A .
- ☐ c. $x = (0, -1, 1)^T$ is an eigenvector of A .

La risposta corretta è: $x = (0, 1, 0)^T$ is an eigenvector of A .

If

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☐ a. $x = (0, 0)^T$ is an eigenvector of A .
- ☒ b. $x = (1, 0)^T$ is an eigenvector of A .
- ☐ c. $x = (1, 1)^T$ is an eigenvector of A .



La risposta corretta è: $x = (1, 0)^T$ is an eigenvector of A .

If

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☒ a. $\lambda = 2$ is the eigenvalue associated with the eigenvector $x = (1, 0)^T$.
- ☐ b. $\lambda = 2$ is the eigenvalue associated with the eigenvector $x = (0, 1)^T$.
- ☐ c. $\lambda = 1$ is the eigenvalue associated with the eigenvector $x = (1, 0)^T$.



La risposta corretta è: $\lambda = 2$ is the eigenvalue associated with the eigenvector $x = (1, 0)^T$.

If $A \in \mathbb{R}^{n \times n}$, $x \in \mathbb{R}^n$ and

$$Ax = \lambda x$$

For $\lambda \in \mathbb{R}$, then:

Scegli un'alternativa:

- ☐ a. For any $c \in \mathbb{R}$, $c \neq 0$, cx is an eigenvector of A .
- ☐ b. cx is an eigenvector of A if and only if $c = 1$.
- ☐ c. None of the above.

La risposta corretta è: For any $c \in \mathbb{R}$, $c \neq 0$, cx is an eigenvector of A .

If

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☐ a. $\lambda = 2$ is the eigenvalue associated with the eigenvector $x = (0, 1, 0)^T$.
- ☐ b. $\lambda = -1$ is the eigenvalue associated with the eigenvector $x = (0, 0, 1)^T$.
- ☒ c. $\lambda = 1$ is the eigenvalue associated with the eigenvector $x = (1, 0, 0)^T$.

✗

La risposta corretta è: $\lambda = -1$ is the eigenvalue associated with the eigenvector $x = (0, 0, 1)^T$.

In $\mathcal{F}(10, 2, -2, 2)$, if $x = \pi$, $y = e$, and $z = fl(x) - fl(y)$, then:

Scegli un'alternativa:

- ☐ a. $fl(z) = 0.43 \times 10^1$.
- ☐ b. $fl(z) = 0.44 \times 10^1$.
- ☒ c. $fl(z) = 0.40 \times 10^1$.



La risposta corretta è: $fl(z) = 0.40 \times 10^1$.

In $\mathcal{F}(10, 2, -2, 2)$, if $x = \pi$, $y = e$, and $z = fl(x) * fl(y)$, then:

Scegli un'alternativa:

- ☒ a. $fl(z) = 0.84 \times 10^1$.
- ☐ b. $fl(z) = 0.0837 \times 10^2$.
- ☐ c. $fl(z) = 0.837 \times 10^1$.



La risposta corretta è: $fl(z) = 0.84 \times 10^1$.

In $\mathcal{F}(10, 6, -3, 3)$, if $x = 192.403$, $y = 0.635782$, and $z = fl(x) + fl(y)$, then:

Scegli un'alternativa:

- ☒ a. $fl(z) = 0.193039 \times 10^3$.
- ☐ b. $fl(z) = 0.193038 \times 10^3$.
- ☐ c. $fl(z) = 0.193038782 \times 10^3$.



La risposta corretta è: $fl(z) = 0.193039 \times 10^3$.

In $\mathcal{F}(10, 2, -2, 2)$, if $x = \pi$, $y = e$, and $z = fl(x) + fl(y)$, then:

Scegli un'alternativa:

- ☐ a. $fl(z) = 0.585 \times 10^1$.
- ☐ b. $fl(z) = 0.58 \times 10^1$.
- ☒ c. $fl(z) = 0.59 \times 10^1$.



La risposta corretta è: $fl(z) = 0.58 \times 10^1$.

If $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $f \in \mathcal{C}^1(\mathbb{R}^n)$, then x^* is a minimum point if and only if:

Scegli un'alternativa:

- ☐ a. $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive semi-definite.
- ☒ b. $\nabla f(x^*) = 0$.
- ☐ c. $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive definite.



La risposta corretta è: $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive definite.

Gradient descent methods:

Scegli un'alternativa:

- ☐ a. If α is suitable chosen, $f \in C^1$, for any x_0 , always converges to a stationary point of $f(x)$.
- ☒ b. If α is suitable chosen, $f \in C^1$, for any x_0 , always converges to a minimum of $f(x)$.
- ☐ c. Always converges to a minimum of $f(x)$.



La risposta corretta è: If α is suitable chosen, $f \in C^1$, for any x_0 , always converges to a stationary point of $f(x)$.

Gradient descent methods solves the optimization problem

$$\min_x f(x)$$

By:

Scegli un'alternativa:

- ☒ a. Generating a sequence $\{x_k\}_k$ such that, given x_0 , computes $x_{k+1} = x_k - \alpha \nabla f(x_k)$ for $\alpha > 0$ step-size.
- ☐ b. Generating a sequence $\{x_k\}_k$ such that, given x_0 , computes $x_{k+1} = x_k + \alpha \nabla f(x_k)$ for $\alpha > 0$ step-size.
- ☐ c. Generating a sequence $\{x_k\}_k$ such that, given x_0 , computes $x_{k+1} = x_k - \alpha \nabla f(x_k)$ for $\alpha \neq 0$ step-size.



La risposta corretta è: Generating a sequence $\{x_k\}_k$ such that, given x_0 , computes $x_{k+1} = x_k - \alpha \nabla f(x_k)$ for $\alpha > 0$ step-size.

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1^2 + x_2^2$, then if the initial guess for a gradient descent iteration is $x^{(0)} = (1, 1)^T$ and $\alpha > 0$, then $|f(x^{(1)})| < |f(x^{(0)})|$ if:

Scegli un'alternativa:

- ☐ a. $0 < \alpha < 1$.
- ☐ b. $\alpha > 0$.
- ☒ c. $\alpha > \frac{1}{2}$.



La risposta corretta è: $0 < \alpha < 1$.

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = e^{x_1} + x_2^2$, then if the initial guess for a gradient descent iteration is $x^{(0)} = (0, 0)^T$ and $\alpha > 0$, then $|f(x^{(1)})| < |f(x^{(0)})|$ if:

Scegli un'alternativa:

- ☐ a. $\alpha > \frac{1}{2}$.
- ☐ b. $\alpha > 0$.
- ☒ c. $0 < \alpha < 1$.



La risposta corretta è: $\alpha > 0$.

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1 e^{x_2}$, then if the initial guess for a gradient descent iteration is $x^{(0)} = (1, 1)^T$ and $\alpha = \frac{1}{2}$, then:

Scegli un'alternativa:

- ☒ a. $x^{(1)} = (1 - \frac{e}{2}, 1 - \frac{e}{2})^T$.
- ☐ b. $x^{(1)} = (1 + \frac{e}{2}, 1 + \frac{e}{2})^T$.
- ☐ c. $x^{(1)} = (\frac{1}{2} - \frac{e}{2}, \frac{1}{2} - \frac{e}{2})^T$.



La risposta corretta è: $x^{(1)} = (1 - \frac{e}{2}, 1 - \frac{e}{2})^T$.

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1 e^{x_2}$, then if the initial guess for a gradient descent iteration is $x^{(0)} = (0, 0)^T$ and $\alpha = 1$, then:

Scegli un'alternativa:

- ☐ a. $x^{(1)} = (1, 0)^T$.
- ☒ b. $x^{(1)} = (-1, 0)^T$.
- ☐ c. $x^{(1)} = (0, 0)^T$.



La risposta corretta è: $x^{(1)} = (-1, 0)^T$.

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = x_1^2 + x_2^2$, then if the initial guess for a gradient descent iteration is $x^{(0)} = (1, 1)^T$ and $\alpha > 0$, then:

Scegli un'alternativa:

- ☒ a. $x^{(1)} = (1 - 2\alpha, 1 - 2\alpha)^T$.
- ☐ b. $x^{(1)} = (1 - \alpha, 1 - \alpha)^T$.
- ☐ c. $x^{(1)} = (1 + 2\alpha, 1 + 2\alpha)^T$.



La risposta corretta è: $x^{(1)} = (1 - 2\alpha, 1 - 2\alpha)^T$.

If $f : \mathbb{R}^2 \rightarrow \mathbb{R}$, $f(x_1, x_2) = e^{x_1} + x_2^2$, then if the initial guess for a gradient descent iteration is $x^{(0)} = (0, 0)^T$ and $\alpha > 0$, then:

Scegli un'alternativa:

- ☐ a. $x^{(1)} = (-\alpha, 0)^T$.
- ☐ b. $x^{(1)} = (0, 0)^T$.
- ☐ c. $x^{(1)} = (-\alpha, 2)^T$.

La risposta corretta è: $x^{(1)} = (-\alpha, 0)^T$.

For Standard IEEE, double precision representation is:

Scegli un'alternativa:

- ☐ a. $\mathcal{F}(2, 64, -1024, 1023)$.
- ☐ b. None of the above.
- ☒ c. $\mathcal{F}(2, 53, -1024, 1023)$.



La risposta corretta è: $\mathcal{F}(2, 53, -1024, 1023)$.

For Standard IEEE, single precision representation is:

Scegli un'alternativa:

- ☐ a. $\mathcal{F}(2, 24, -128, 127)$.
- ☐ b. None of the above.
- ☒ c. $\mathcal{F}(2, 32, -128, 127)$.



La risposta corretta è: $\mathcal{F}(2, 24, -128, 127)$.

Given two independent random variables X and Y , then:

Scegli un'alternativa:

- ☐ a. $p(x) = p(y)$
- ☒ b. $p(y) = p(y|x)$
- ☐ c. $p(x|y) = p(y)$



La risposta corretta è: $p(y) = p(y|x)$.

Given two random variables X and Y , Bayes Theorem implies that:

Scegli un'alternativa:

- ☐ a. $p(x) = p(y)p(y|x) / p(y|x)$.
- ☒ b. $p(x) = p(x|y)p(y|x) / p(y)$.
- ☐ c. $p(y) = p(y|x)p(x) / p(x|y)$.



La risposta corretta è: $p(y) = p(y|x)p(x) / p(x|y)$.

If $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $f(x) = \|Ax - b\|_2^2$ for $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, then:

Scegli un'alternativa:

- ☒ a. $\nabla f(x) = 2A^T(Ax - b)$.
- ☐ b. $\nabla f(x) = A^T(Ax - b)$.
- ☐ c. $\nabla f(x) = A(A^T x - b)$.



La risposta corretta è: $\nabla f(x) = 2A^T(Ax - b)$.

If $f : \mathbb{R}^n \rightarrow \mathbb{R}, f(x) = \|Ax - b\|_2^2$ for $A \in \mathbb{R}^{m \times n}, b \in \mathbb{R}^m$, then the solution of $\nabla f(x) = 0$ is:

Scegli un'alternativa:

- ☐ a. $A^T Ax = b$
- ☐ b. $A^T Ax = A^T x$
- ☒ c. $Ax = b.$



La risposta corretta è: $A^T Ax = A^T x$

The machine precision ϵ can be defined as:

Select one:

- ☐ a. The smallest number ϵ such that $fl(1 + \epsilon) = 1$.
- ☒ b. The smallest number ϵ such that $fl(1 + \epsilon) > 1$.
- ☐ c. None of the above.



The correct answer is: The smallest number ϵ such that $fl(1 + \epsilon) > 1$.

Given two random variables X and Y such that $p(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x}$ and $p(y|x) = ce^{-|y-ax|}$, then the MAP reads:

Scegli un'alternativa:

- ☐ a. $x^* = \arg \min_x |y - ax| + \frac{1}{2}x^2$.
- ☒ b. $x^* = \arg \min_x |y - ax|$.
- ☐ c. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2$.



La risposta corretta è: $x^* = \arg \min_x |y - ax|$.

Given two random variables X and Y such that $p(x) = ce^{-|x|}$ and $p(y|x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(y-ax)^2}$, then the MAP reads:

Scegli un'alternativa:

- ☒ a. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2 + |x|.$
- ☐ b. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2 + \frac{1}{2}x^2.$
- ☐ c. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2.$



La risposta corretta è: $x^* = \arg \min_x \frac{1}{2}(y - ax)^2 + |x|.$

Given two random variables X and Y such that $p(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}x^2}$ and $p(y|x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(y-ax)^2}$, then the MAP reads:

Scegli un'alternativa:

- ☐ a. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2 + \frac{1}{2}x^2$.
- ☐ b. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2 + x^2$.
- ☐ c. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2$.

La risposta corretta è: $x^* = \arg \min_x \frac{1}{2}(y - ax)^2 + \frac{1}{2}x^2$.

If

$$A = \begin{bmatrix} 2 & 1 & -2 \\ -1 & 0 & 1 \\ -1 & 2 & 1 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☐ a. A is orthogonal.
- ☒ b. None of the above.
- ☐ c. A is symmetric and definite positive.



La risposta corretta è: None of the above.

If

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☒ a. A is symmetric and positive definite.
- ☐ b. A is non-symmetric and not positive definite.
- ☐ c. A is symmetric but not positive definite.



La risposta corretta è: A is symmetric and positive definite.

If

$$A = \begin{bmatrix} \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ -\frac{2}{3} & \frac{2}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{2}{3} & -\frac{2}{3} \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☐ a. A is symmetric and definite positive.
- ☐ b. A is symmetric but not definite positive.
- ☒ c. A is orthogonal.



La risposta corretta è: A is orthogonal.

If

$$A = \begin{bmatrix} 2 & 2 & -1 \\ 2 & 0 & 2 \\ -1 & 2 & 3 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☒ a. A is symmetric but not definite positive.
- ☐ b. A is symmetric and definite positive.
- ☐ c. A is orthogonal.



La risposta corretta è: A is symmetric but not definite positive.

If

$$A = \begin{bmatrix} -1 & 1 \\ 0 & 3 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☐ a. A is symmetric and positive definite.
- ☒ b. A is non-symmetric and not positive definite.
- ☐ c. A is symmetric but not positive definite.



La risposta corretta è: A is non-symmetric and not positive definite.

If

$$A = \begin{bmatrix} 9 & 6 \\ 6 & 5 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☐ a. A is symmetric but not positive definite.
- ☐ b. A is non-symmetric and not positive definite.
- ☒ c. A is symmetric and positive definite.



La risposta corretta è: A is symmetric and positive definite.

If

$$A = \begin{bmatrix} -1 & 0 \\ 0 & 3 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☐ a. A is symmetric and positive definite.
- ☒ b. A is symmetric but not positive definite.
- ☐ c. A is non-symmetric and not positive definite.



La risposta corretta è: A is symmetric but not positive definite.

If

$$A = \begin{bmatrix} 2 & 1 & 2 \\ -2 & 2 & 1 \\ 1 & 2 & -2 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☐ a. A is not orthogonal.
- ☒ b. A is orthogonal.
- ☐ c. A is symmetric but not definite positive.



La risposta corretta è: A is not orthogonal.

If

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 1 & 1 & 0 \\ -1 & 1 & 2 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☐ a. $\text{rank}(A) = 2$.
- ☐ b. $\text{rank}(A) = 1$.
- ☒ c. $\text{rank}(A) = 3$.



La risposta corretta è: $\text{rank}(A) = 3$.

If

$$A = \begin{bmatrix} 0 & 6 & 8 \\ 2 & 4 & 0 \\ 1 & 0 & 8 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☒ a. $\text{rank}(A) = 3$.
- ☐ b. $\text{rank}(A) = 2$.
- ☐ c. $\text{rank}(A) = 1$.



La risposta corretta è: $\text{rank}(A) = 3$.

If

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☒ a. $\text{rank}(A) = 2$.
- ☐ b. $\text{rank}(A) = 1$.
- ☐ c. $\text{rank}(A) = 3$.



La risposta corretta è: $\text{rank}(A) = 2$.

If

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

then:

Scegli un'alternativa:

- ☒ a. $\text{rank}(A) = 3$.
- ☐ b. $\text{rank}(A) = 4$.
- ☐ c. $\text{rank}(A) = 2$.



La risposta corretta è: $\text{rank}(A) = 3$.

If

$$A = \begin{bmatrix} 1 & -1 & 1 \\ 2 & -2 & 2 \\ -1 & 1 & -1 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☐ a. $\text{rank}(A) = 3$.
- ☒ b. $\text{rank}(A) = 1$.
- ☐ c. $\text{rank}(A) = 2$.



La risposta corretta è: $\text{rank}(A) = 1$.

If

$$A = \begin{bmatrix} 0 & 2 & -1 \\ 1 & 1 & 0 \\ 1 & 3 & -1 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☒ a. $\text{rank}(A) = 2$.
- ☐ b. $\text{rank}(A) = 3$.
- ☐ c. $\text{rank}(A) = 1$.



La risposta corretta è: $\text{rank}(A) = 2$.

If

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

then:

Scegli un'alternativa:

- ☒ a. $\text{rank}(A) = 4$.
- ☐ b. $\text{rank}(A) = 2$.
- ☐ c. $\text{rank}(A) = 3$.



La risposta corretta è: $\text{rank}(A) = 4$.

If

$$A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 2 & 1 & 2 \end{bmatrix}$$

Then:

Scegli un'alternativa:

- ☐ a. $\text{rank}(A) = 1$.
- ☐ b. $\text{rank}(A) = 3$.
- ☒ c. $\text{rank}(A) = 2$.



La risposta corretta è: $\text{rank}(A) = 2$.

Given two random variables X and Y such that $p(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x}$ and $p(y|x) = ce^{-|y-ax|}$, then the MLE reads:

Scegli un'alternativa:

- ☐ a. $x^* = \arg \min_x |y - ax| + x^2$.
- ☐ b. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2$.
- ☒ c. $x^* = \arg \min_x |y - ax|$.



La risposta corretta è: $x^* = \arg \min_x |y - ax|$.

Given two random variables X and Y such that $p(x) = ce^{-|x|}$ and $p(y|x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}(y-ax)^2}$, then the MLE reads:

Scegli un'alternativa:

- ☐ a. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2 + |x|$.
- ☐ b. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2 + \frac{1}{2}x^2$.
- ☐ c. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2$.

La risposta corretta è: $x^* = \arg \min_x \frac{1}{2}(y - ax)^2$.

Given two random variables X and Y such that $p(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x}$ and $p(y|x) = ce^{-|y-ax|}$, then the MLE reads:

Scegli un'alternativa:

- ☐ a. $x^* = \arg \min_x |y - ax| + x^2$.
- ☐ b. $x^* = \arg \min_x |y - ax|$.
- ☒ c. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2$.

✗

La risposta corretta è: $x^* = \arg \min_x |y - ax|$.

Given two random variables X and Y such that $p(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$ and $p(y|x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(y-ax)^2}$, then the MLE reads:

Scegli un'alternativa:

- ☐ a. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2 + \frac{1}{2}x^2$.
- ☐ b. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2 + x^2$.
- ☒ c. $x^* = \arg \min_x \frac{1}{2}(y - ax)^2$.

□

La risposta corretta è: $x^* = \arg \min_x \frac{1}{2}(y - ax)^2$.

If

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

then:

Scegli un'alternativa:

- ☐ a. The 2-norm of A is $\|A\|_2 = 1$.
- ☐ b. The 2-norm of A is $\|A\|_2 = 0$.
- ☒ c. The 2-norm of A is $\|A\|_2 = 3$.



La risposta corretta è: The 2-norm of A is $\|A\|_2 = 3$.

If

$$A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 4 \end{bmatrix}$$

then:

Scegli un'alternativa:

- ☐ a. The 2-norm of A is $\|A\|_2 = 2$.
- ☒ b. The 2-norm of A is $\|A\|_2 = 4$.
- ☐ c. The 2-norm of A is $\|A\|_2 = 2$.



La risposta corretta è: The 2-norm of A is $\|A\|_2 = 4$.

If

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

then:

Scegli un'alternativa:

- ☒ a. The 2-norm of A is $\|A\|_2 = 3$.
- ☐ b. The 2-norm of A is $\|A\|_2 = 0$.
- ☐ c. The 2-norm of A is $\|A\|_2 = 3$.



Le risposte corrette sono: The 2-norm of A is $\|A\|_2 = 3$, The 2-norm of A is $\|A\|_2 = 3$.

If A is an $n \times n$ matrix, then

Scegli un'alternativa:

- ☐ a. $\|A\|_1 = \sqrt{\sum_{i=1}^n \sum_{j=1}^n a_{i,j}^2}$.
- ☐ b. $\|A\|_1 = \rho(A^T A)$.
- ☒ c. None of the above.



La risposta corretta è: None of the above.

If A is an $n \times n$ matrix, then

Scegli un'alternativa:

- ☒ a. None of the above.
- ☐ b. $\|A\|_2 = \rho(A^T A)$.
- ☐ c. $\|A\|_2 = \sqrt{\sum_{i=1}^n \sum_{j=1}^n a_{i,j}^2}$.

✖

La risposta corretta è: $\|A\|_2 = \rho(A^T A)$.

If A is an $n \times n$ matrix, then

Scegli un'alternativa:

- ☒ a. $\|A\|_F = \sqrt{\sum_{i=1}^n \sum_{j=1}^n a_{i,j}^2}$.
- ☐ b. None of the above.
- ☐ c. $\|A\|_F = \rho(A^T A)$.



La risposta corretta è: $\|A\|_F = \sqrt{\sum_{i=1}^n \sum_{j=1}^n a_{i,j}^2}$.

If $A \in \mathbb{R}^{m \times n}$, $\|A\|_p = 0$, then:

Scegli un'alternativa:

- ☒ a. $A = 0$.
- ☐ b. $\text{rank}(A) = 0$.
- ☐ c. A can be both equal or not equal to 0.



La risposta corretta è: $A = 0$.

A matrix $A \in \mathbb{R}^{n \times n}$ is orthogonal if:

Scegli un'alternativa:

- ☐ a. $A^{-1}A = I = AA^{-1}$.
- ☒ b. $A^T A = I = AA^T$.
- ☐ c. $A = A^T$.



La risposta corretta è: $A^T A = I = AA^T$.

If $X : \Omega \rightarrow \mathcal{T}$ is a continuous random variable, then a function $p : \mathcal{T} \rightarrow \mathbb{R}_+$ can be the PDF of X if:

Scegli un'alternativa:

- ☒ a. $\int_{\mathcal{T}} p(x)dx = 1.$
- ☐ b. $\int_{\Omega} p(x)dx = 1.$
- ☐ c. $\int_{\mathcal{T}} p(x)dx < \infty.$



La risposta corretta è: $\int_{\mathcal{T}} p(x)dx = 1.$

For a random variable $X : \Omega \rightarrow \mathcal{T}$ with $\mathbb{E}[X] = 0$, it holds:

Scegli un'alternativa:

- ☐ a. $Var(X) = \mathbb{E}[X]$.
- ☒ b. $Var(X) = \mathbb{E}[X^2]$.
- ☐ c. $Var(X) = 0$.



La risposta corretta è: $Var(X) = \mathbb{E}[X^2]$.

If $X : \Omega \rightarrow \mathcal{T}$ is a discrete random variable, then a function $f_X : \mathcal{T} \rightarrow [0, 1]$ can be the PDF of X if:

Scegli un'alternativa:

- ☐ a. $\sum_{i \in \Omega} f_X(i) = 1.$
- ☒ b. $\int_{\Omega} f_X(x) dx = 1.$
- ☐ c. $\sum_{i \in \mathcal{T}} f_X(i) = 1.$



La risposta corretta è: $\sum_{i \in \mathcal{T}} f_X(i) = 1.$

Given a discrete random variable $X : \Omega \rightarrow \mathcal{T}$, with $\mathcal{T} = \{1, 2, \dots, 6\}$, and $f_X = \{\frac{1}{6}, \frac{1}{6}, \dots, \frac{1}{6}\}$, then:

Scegli un'alternativa:

- ☐ a. $\mathbb{E}[X] = 21$.
- ☐ b. $\mathbb{E}[X] = 3.5$.
- ☒ c. $\mathbb{E}[X] = \frac{1}{6}$.



La risposta corretta è: $\mathbb{E}[X] = 3.5$.

Given a continuous random variable $X : \Omega \rightarrow \mathcal{T}$, with $\mathcal{T} = [0, 1]$, and $p(x) = 3x^2$ its PDF, then:

Scegli un'alternativa:

- ☐ a. $\mathbb{E}[X] = 2$.
- ☐ b. $\mathbb{E}[X] = 3$.
- ☒ c. $\mathbb{E}[X] = \frac{3}{4}$.



La risposta corretta è: $\mathbb{E}[X] = \frac{3}{4}$.

Given a continuous random variable $X : \Omega \rightarrow \mathcal{T}$, with $\mathcal{T} = [0, 1]$, and $p(x) = 2x$ its PDF, then:

Scegli un'alternativa:

- ☐ a. $\mathbb{E}[X] = \frac{2}{3}$.
- ☐ b. $\mathbb{E}[X] = 2$.
- ☐ c. $\mathbb{E}[X] = 1$.

La risposta corretta è: $\mathbb{E}[X] = \frac{2}{3}$.

If $X : \Omega \rightarrow \mathcal{T}$ is a continuous random variable with PDG $p : \mathcal{T} \rightarrow \mathbb{R}_+$, then:

Scegli un'alternativa:

- ☐ a. $\mathbb{E}[X] = \int_{\mathcal{T}} p(x)dx.$
- ☒ b. $\mathbb{E}[X] = \int_{\Omega} xp(x)dx.$
- ☐ c. $\mathbb{E}[X] = \int_{\mathcal{T}} xp(x)dx.$

✖

La risposta corretta è: $\mathbb{E}[X] = \int_{\mathcal{T}} xp(x)dx.$

If $X : \Omega \rightarrow \mathcal{T}$ is a continuous random variable, its Probability Density Function (PDF) $p_X(x)$ is defined to be:

Scegli un'alternativa:

- ☐ a. $P(X = x) = p_X(x)$.
- ☐ b. $P(X = x) = \int x p_X(x) dx$.
- ☒ c. $P(X \in A) = \int_A p_X(x) dx$.



La risposta corretta è: $P(X \in A) = \int_A p_X(x) dx$.

If $X : \Omega \rightarrow \mathcal{T}$ is a continuous random variable, its Probability Density Function (PDF) $p_X(x)$ is:

Scegli un'alternativa:

- ☐ a. A function $p_X : \mathcal{T} \rightarrow \mathbb{R}_+$.
- ☐ b. A function $p_X : \mathcal{T} \rightarrow [0, 1]$.
- ☐ c. A function $p_X : \Omega \rightarrow [0, 1]$.

La risposta corretta è: A function $p_X : \mathcal{T} \rightarrow \mathbb{R}_+$.

Given two discrete random variable $X_1 : \Omega \rightarrow \mathcal{T}$, $X_2 : \Omega \rightarrow \mathcal{T}$ with $\mathcal{T} = \{1, 2, 3\}$, and $f_{X_1} = \{\frac{1}{2}, \frac{1}{6}, \frac{1}{3}\}$, $f_{X_2} = \{\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\}$ their PMF, then:

Scegli un'alternativa:

- ☐ a. $\mathbb{E}[X_1] > \mathbb{E}[X_2]$.
- ☐ b. $\mathbb{E}[X_1] = \mathbb{E}[X_2]$.
- ☒ c. $\mathbb{E}[X_1] < \mathbb{E}[X_2]$.



La risposta corretta è: $\mathbb{E}[X_1] < \mathbb{E}[X_2]$.

Given two discrete random variable $X_1 : \Omega \rightarrow \mathcal{T}$, $X_2 : \Omega \rightarrow \mathcal{T}$ with $\mathcal{T} = \{1, 2, 3\}$, and $f_{X_1} = \{\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\}$, $f_{X_2} = \{\frac{1}{2}, \frac{1}{6}, \frac{1}{3}\}$ their PMF, then:

Scegli un'alternativa:

- ☐ a. $\mathbb{E}[X_1] < \mathbb{E}[X_2]$.
- ☐ b. $\mathbb{E}[X_1] = \mathbb{E}[X_2]$.
- ☒ c. $\mathbb{E}[X_1] > \mathbb{E}[X_2]$.



La risposta corretta è: $\mathbb{E}[X_1] > \mathbb{E}[X_2]$.

If $X : \Omega \rightarrow \mathcal{T}$ is a discrete random variable, its Probability Mass Function (PMF) f_x is:

Scegli un'alternativa:

- ☐ a. $f_X(x) = \int P(x)dx.$
- ☐ b. $f_X(x) = P(X \in x).$
- ☒ c. $f_X(x) = P(X = x).$



La risposta corretta è: $f_X(x) = P(X = x).$

If $X : \Omega \rightarrow \mathcal{T}$ is a discrete random variable with PMG $f_X : \mathcal{T} \rightarrow [0, 1]$, then:

Scegli un'alternativa:

☐ a. $\mathbb{E}[X] = \sum_{i \in \mathcal{T}} i.$

☒ b. $\mathbb{E}[X] = \sum_{i \in \mathcal{T}} i f_X(i).$



Vai a...

La risposta corretta è: $\mathbb{E}[X] = \sum_{i \in \mathcal{T}} i f_X(i).$

Given a discrete random variable $X : \Omega \rightarrow \mathcal{T}$, with $\mathcal{T} = \{1, 2, 3\}$, and $f_X = \{\frac{1}{2}, \frac{1}{6}, \frac{1}{3}\}$ its PMF, then:

Scegli un'alternativa:

- ☐ a. $\mathbb{E}[X] = 6$.
- ☐ b. $\mathbb{E}[X] = 2$.
- ☒ c. $\mathbb{E}[X] = \frac{11}{6}$.



La risposta corretta è: $\mathbb{E}[X] = \frac{11}{6}$.

Given a discrete random variable $X : \Omega \rightarrow \mathcal{T}$, with $\mathcal{T} = \{1, 2, 3\}$, and $f_X = \{\frac{1}{6}, \frac{1}{3}, \frac{1}{2}\}$ its PMF, then:

Scegli un'alternativa:

- ☒ a. $\mathbb{E}[X] = 6$.
- ☐ b. $\mathbb{E}[X] = 2$.
- ☐ c. $\mathbb{E}[X] = \frac{13}{6}$.



La risposta corretta è: $\mathbb{E}[X] = \frac{13}{6}$.

If $X : \Omega \rightarrow \mathcal{T}$ is a discrete random variable, its Probability Mass Function (PMF) f_x is:

Select one:

- ☒ a. A function $f_X : \mathcal{T} \rightarrow [0, 1]$.
- ☐ b. A function $f_X : \Omega \rightarrow [0, 1]$.
- ☐ c. A function $f_X : \mathcal{T} \rightarrow \mathbb{R}$.



The correct answer is: A function $f_X : \mathcal{T} \rightarrow [0, 1]$.

Given two discrete random variable $X_1 : \Omega \rightarrow \mathcal{T}$, $X_2 : \Omega \rightarrow \mathcal{T}$ with $\mathcal{T} = \{1, 2, 3\}$, and $f_{X_1} = \{\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\}$, $f_{X_2} = \{\frac{1}{2}, \frac{1}{6}, \frac{1}{3}\}$ their PMF, then:

Scegli un'alternativa:

- ☒ a. $\mathbb{E}[X_1] < \mathbb{E}[X_2]$.
- ☐ b. $\mathbb{E}[X_1] = \mathbb{E}[X_2]$.
- ☐ c. $\mathbb{E}[X_1] > \mathbb{E}[X_2]$.



La risposta corretta è: $\mathbb{E}[X_1] > \mathbb{E}[X_2]$.

The precision is:

Scegli un'alternativa:

- ☐ a. None of the above.
- ☒ b. The number of digits with which a number is expressed.
- ☐ c. The number of correct significant digits in approximating some quantity.



La risposta corretta è: The number of digits with which a number is expressed.

A random variable X is:

Scegli un'alternativa:

- ☐ a. A function $X : \Omega \rightarrow \mathcal{T}$.
- ☒ b. A variable that returns random elements with known probability.
- ☐ c. A set that contains the possible outcomes of the experiment.



La risposta corretta è: A function $X : \Omega \rightarrow \mathcal{T}$.

If Ω is the sample space, $\mathcal{A}$ is the event space and $\mathcal{T}$ is a subset of $\mathbb{R}$, a random variable X is:

Scegli un'alternativa:

- ☐ a. A function $X : \Omega \rightarrow \mathcal{A}$.
- ☐ b. A function $X : \mathcal{A} \rightarrow \mathcal{T}$.
- ☒ c. A function $X : \Omega \rightarrow \mathcal{T}$.



La risposta corretta è: A function $X : \Omega \rightarrow \mathcal{T}$.

In normalized scientific notation and base $\beta = 10$, if $x = 2.71$, then:

Scegli un'alternativa:

- ☒ a. The mantissa of x is 0.271 and the exponential part is 10^1 .
- ☐ b. The mantissa of x is 2.71 and the exponential part is 10^0 .
- ☐ c. None of the above.



La risposta corretta è: The mantissa of x is 0.271 and the exponential part is 10^1 .

If $A = U\Sigma V^T$ is the SVD decomposition of $A \in \mathbb{R}^{m \times n}$, then a dyade $A_i = u_i v_i^T$ of A is:

Scegli un'alternativa:

- ☐ a. None of the above.
- ☐ b. A vector of length mn that express some properties of A .
- ☒ c. A rank-1 matrix of dimension $m \times n$.



La risposta corretta è: A rank-1 matrix of dimension $m \times n$.

If $A = U\Sigma V^T$ is the SVD decomposition of an $m \times n$ matrix A , then:

Scegli un'alternativa:

- ☐ a. $A^T A = V^T \Sigma^2 V$.
- ☐ b. $A^T A = U \Sigma^2 U^T$.
- ☒ c. $A^T A = V \Sigma^2 V^T$.



La risposta corretta è: $A^T A = V \Sigma^2 V^T$.

If $A = U\Sigma V^T$ is the SVD decomposition of an $m \times n$ matrix A , then:

Scegli un'alternativa:

- ☐ a. The rows of V^T are eigenvectors of AA^T .
- ☒ b. The columns of U are eigenvectors of AA^T .
- ☐ c. None of the above



La risposta corretta è: The columns of U are eigenvectors of AA^T .

If $A = U\Sigma V^T$ is the SVD decomposition of an $m \times n$ matrix A , then:

Scegli un'alternativa:

- ☐ a. The rows of V^T are eigenvectors of $A^T A$.
- ☒ b. None of the above
- ☐ c. The columns of U are eigenvectors of $A^T A$.



La risposta corretta è: The columns of U are eigenvectors of $A^T A$.

Given a matrix $A \in \mathbb{R}^{m \times n}$, $m > n$, with $r = \text{rank}(A)$, then:

Scegli un'alternativa:

- ☐ a. None of the above.
- ☒ b. It is always possible to write A as $U\Sigma V^T$, where $\Sigma \in \mathbb{R}^{m \times n}$ is diagonal, $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$ are orthogonal. ✓
- ☐ c. It is possible to write A as $U\Sigma V^T$, where $\Sigma \in \mathbb{R}^{m \times n}$ is diagonal, $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$ are orthogonal if and only if $\text{rank}(A) = n$.

La risposta corretta è: It is always possible to write A as $U\Sigma V^T$, where $\Sigma \in \mathbb{R}^{m \times n}$ is diagonal, $U \in \mathbb{R}^{m \times m}$, $V \in \mathbb{R}^{n \times n}$ are orthogonal.

If $A = U\Sigma V^T$ is the SVD decomposition of an $m \times n$ matrix A , then:

Scegli un'alternativa:

- ☐ a. $AA^T = U^T \Sigma^2 U$.
- ☒ b. $AA^T = U \Sigma^2 U^T$.
- ☐ c. $AA^T = U \Sigma U^T$.



La risposta corretta è: $AA^T = U \Sigma^2 U^T$.

If $A = U\Sigma V^T$ is the SVD decomposition of an $m \times n$ matrix A , then:

Scegli un'alternativa:

- ☒ a. The elements of the diagonal matrix Σ are the singular values of A , in decreasing order.
- ☐ b. The elements of the diagonal matrix Σ are the eigenvalues of A , in decreasing order.
- ☐ c. None of the above.



La risposta corretta è: The elements of the diagonal matrix Σ are the singular values of A , in decreasing order.

If $A = U\Sigma V^T$ is the SVD decomposition of an $m \times n$ matrix A , then:

Scegli un'alternativa:

- ☐ a. The singular values σ_i of A are $\sigma_i = \sqrt{\lambda_i(A^T A)}$ where $\lambda_i(A^T A)$ are the eigenvalues of $A^T A$.
- ☐ b. None of the above
- ☒ c. The singular values σ_i of A are $\sigma_i = \lambda_i(A^T A)$ where $\lambda_i(A^T A)$ are the eigenvalues of $A^T A$.



La risposta corretta è: The singular values σ_i of A are $\sigma_i = \sqrt{\lambda_i(A^T A)}$ where $\lambda_i(A^T A)$ are the eigenvalues of $A^T A$.

If $A = U\Sigma V^T$ is the SVD decomposition of $A \in \mathbb{R}^{m \times n}$, then the rank k approximation of A is:

Scegli un'alternativa:

- ☐ a. $\hat{A}(k) = \sum_{i=1}^n \sigma_i A_i$, where $A_i = u_i v_i^T$ is a dyade.
- ☐ b. $\hat{A}(k) = \sum_{i=1}^k A_i$, where $A_i = u_i v_i^T$ is a dyade.
- ☒ c. $\hat{A}(k) = \sum_{i=1}^k \sigma_i A_i$, where $A_i = u_i v_i^T$ is a dyade.



La risposta corretta è: $\hat{A}(k) = \sum_{i=1}^k \sigma_i A_i$, where $A_i = u_i v_i^T$ is a dyade.

If $A = U\Sigma V^T$ is the SVD decomposition of an $m \times n$ matrix A , then:

Scegli un'alternativa:

- ☐ a. The elements on the diagonal of Σ are strictly positive.
- ☐ b. None of the above.
- ☒ c. The elements of the diagonal matrix Σ are non-negative.



La risposta corretta è: The elements of the diagonal matrix Σ are non-negative.

If $A = U\Sigma V^T$ is the SVD decomposition of $A \in \mathbb{R}^{m \times n}$, then its rank k approximation of $\hat{A}(k)$ satisfies:

Select one:

- ☒ a. $\hat{A}(k) = \arg \min_{rk(B)=k} \|A - B\|_2.$
- ☐ b. $\hat{A}(k) = \arg \min_{rk(B)=k} \|A - B\|_F.$
- ☐ c. $\hat{A}(k) = \sigma_{k+1}.$



The correct answer is: $\hat{A}(k) = \arg \min_{rk(B)=k} \|A - B\|_2.$

If $A \in \mathbb{R}^{m \times n}$, then:

Scegli un'alternativa:

- ☒ a. $A^T A$ is symmetric but not necessarily positive definite.
- ☐ b. $A^T A$ is always symmetric and positive definite.
- ☐ c. It depends on A .

✗

La risposta corretta è: $A^T A$ is always symmetric and positive definite.

For a random variable $X : \Omega \rightarrow \mathcal{T}$, its variance is defined as:

Scegli un'alternativa:

- ☐ a. $Var(X) = \mathbb{E}[X - \mathbb{E}[X]]$.
- ☒ b. $Var(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$.
- ☐ c. $Var(X) = \mathbb{E}[X^2 - \mathbb{E}[X]^2]$.



La risposta corretta è: $Var(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$.

For a random variable $X : \Omega \rightarrow \mathcal{T}$, it holds:

Scegli un'alternativa:

- ☒ a. $Var(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.
- ☐ b. $Var(X) = \mathbb{E}[X^2] + \mathbb{E}[X]^2$.
- ☐ c. $Var(X) = \mathbb{E}[X]^2 - \mathbb{E}[X^2]$.



La risposta corretta è: $Var(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.

For a random variable $X : \Omega \rightarrow \mathcal{T}$ with $\mathbb{E}[X] = 0$, it holds:

Scegli un'alternativa:

- ☒ a. $\text{Var}(X) = \mathbb{E}[X^2]$.
- ☐ b. $\text{Var}(X) = 0$.
- ☐ c. $\text{Var}(X) = \mathbb{E}[X]$.



La risposta corretta è: $\text{Var}(X) = \mathbb{E}[X^2]$.

If vector $v = (10^6, 0)^T$ is approximated by vector $\tilde{v} = (999996, 1)^T$, then in $\|\cdot\|_1$ the relative error between v and $\tilde{v}$ is:

Scegli un'alternativa:

- ☒ a. $4 \cdot 10^{-6}$.
- ☐ b. None of the above.
- ☐ c. $5 \cdot 10^{-6}$.



La risposta corretta è: $5 \cdot 10^{-6}$.

If vector $v = (10^6, 0)^T$ is approximated by vector $\tilde{v} = (999996, 1)^T$, then in $\|\cdot\|_2$ the relative error between v and $\tilde{v}$ is:

Scegli un'alternativa:

- ☒ a. $\sqrt{17} \cdot 10^{-6}$.
- ☐ b. $4 \cdot 10^{-6}$.
- ☐ c. None of the above.



La risposta corretta è: $\sqrt{17} \cdot 10^{-6}$.

If vector $v = (1, 0)^T$ is approximated by vector $\tilde{v} = (1.1, 0.1)^T$, then in $\|\cdot\|_1$ the relative error between v and $\tilde{v}$ is:

Scegli un'alternativa:

- ☐ a. 0.1.
- ☒ b. 0.2.
- ☐ c. None of the above.



La risposta corretta è: 0.2.

If vector $v = (10^6, 1)^T$ is approximated by vector $\tilde{v} = (999996, 1)^T$, then in $\|\cdot\|_\infty$ the relative error between v and $\tilde{v}$ is:

Scegli un'alternativa:

- ☒ a. 4.
- ☐ b. $4 \cdot 10^{-6}$.
- ☐ c. None of the above.



La risposta corretta è: $4 \cdot 10^{-6}$.

Given $x = (1, -1)^T$ and $y = (2, 1)^T$, then:

Scegli un'alternativa:

- ☒ a. None of the above.
- ☐ b. $x \perp y$.
- ☐ c. $x \parallel y$.



La risposta corretta è: None of the above.

Given $x = (1, -1)^T$ and $y = (1, 1)^T$, then:

Scegli un'alternativa:

- ☒ a. $x \perp y$.
- ☐ b. None of the above.
- ☐ c. $x \parallel y$.



La risposta corretta è: $x \perp y$.